

Milad Heydari

EDUCATION

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Georgia Institute of Technology

BS in Computer Science - AI, Math concentrations

3.86 GPA

MS in Machine Learning

4.0 GPA

Courses: Deep Learning, Deep Reinforcement Learning, Vision-Language Models, Natural Language, OS

ML Systems, HPC & GPU Programming, Computer Vision, Robotics, Statistical ML, Machine Learning

EXPERIENCE

Databricks

Software Engineer — Photon, low latency + performance pod

Feb 2025 -

San Francisco, CA

Databricks

Software Engineering Intern — Photon (C++ version of Spark / [paper link](#)) [mentor](#)

May - Aug 2024

San Francisco, CA

- Designed novel vectorized broadcast join algorithm that was more efficient than SOTA for sparse batches, setting a benchmarking record (85%+ speedup) & saving many millions in compute to date. O(n) speedup.
- Independently found above optimization opportunity while reading code/paper. Outside of assigned scope, but proposed new project / redesign leveraging SIMD, CUDA kernels, and zero-copying to accelerate data loading.
- Implemented memory coalescing during batch processing of joins, improving throughput by 32x.

Microsoft

Software Engineering (Applied Science / ML) Intern — Azure Hyperscale Networking [mentor](#)

May - Aug 2023

Redmond, WA

- Lead project to model distributed system as real-time updating knowledge graph with devices (nodes), links (edges), and anomaly events (temporal nodes): enabling modeling spatial relationships between anomalies
- Built API that queried network/graph to detect & cluster anomalies. Fed data to LLM using RAG to diagnose issues e.g: using packet loss between datacenters to locate bad cables. 22% drop in avg time to diagnose issues

Cockroach Labs

Software Engineering Intern — Cloud Infrastructure & Distributed Systems

Jan - May 2023

New York, NY

- Shipped service to detect underused Kubernetes clusters and mitigate resource waste by enforcing a resource virtualized “sleep” state and predicatively scaling compute based on demand. Saving \$70,000+ annually
- Enabled multi-cloud data querying/hosting via upgrading consensus algorithm for KV / database transactions

Microsoft

Software Engineering Intern — Azure Networking, Applied Sciences Group under [Jitu Padhye](#)

May - Aug 2022

Redmond, WA

- Built backtesting framework for ML models. Allow for A/B and unit testing different models and hyperparams
- Designed architecture/model for time series anomaly detection on streamed data. Performs real time signal processing on a distributed system’s network data and triggering mitigation. Prevented 3 incidents in 1st week

Microsoft

Data & Applied Science Intern — Machine Learning Product Integration

May - Aug 2021

Cambridge, MA

- Used auto-encoders to improve model efficiency and preserve privacy of hospital data during federated learning
- Worked with client, Novant Health, to develop custom NLP/ML models and data pipelines to process language data (encoder model) and monitor metrics (RNNs) in diagnoses recommendation/search tool for doctors

RR AI

Robotics & Deep Learning Research Intern — Computer Vision on Autonomous Vehicles

May - Aug 2020

Washington D.C

- GANs / programmatic synthetic data generation to supplement training of vision models
- During AV fleet movement, projected position of other vehicles into camera view of each vehicle - using positions to test/calibrate/train predicted positions from onboard vision model and assist with navigation/SLAM

PROJECTS

Distributed Training: locking-free distributed SGD. Scaling Hogwild without atomicity guarantees

Inference Scheduler: vLLM fork with virtual token counter scheduler, batching {prefills, decodes} without stalling

LLM reasoning finetuning: improving Quiet-sTaR & sTaR chain of thought reasoning using DPO training

3D VLM editor: adding/removing objects to Gaussian splats using language. Joint CLIP/grouping training

RL autograd framework: backprop with support for randomized variables and stochastic/RL problems

Robust RLHF in LLM Training: achieving stable reward despite variance in human feedback

SKILLS

Languages: Python, C++, C, Golang, C#, Java, Scala, JavaScript, SQL

Tools: PyTorch, CUDA, Kubernetes, MPI, OpenMP, Spark, OpenCV, NumPy, Pandas, gRPC, HuggingFace, Wandb, LangChain, AWS, Azure, GCP, Cursor